SAT Math Final Revision Set 2

76 questions · 2 hours · Answers and solutions

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (13 items)

1. A-317 · Hard · Multiple choice · **No calculator needed**

The function f is linear, $f(5) + f(-5) = 14$, and $f(2) = 13$. What is the value of $f(-2)$?

A) -13

B) 1

C) -1

D) 7

Answer: (B) 1

For $f(x) = mx + b$: $f(c) + f(-c) = 2b$ for every c. So $2b = 14$ and $f(2) + f(-2) = 14$, giving $f(-2) = 14 - 13 = 1$. ($m = 3$, $b = 7$.)

Trap: Students treat f as odd and answer -13, or try to find m and b from one point.

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2. A-320 · Hard · Multiple choice · **No calculator needed**

In the equation $ax + by = c$, a, b and c are constants with $a > 0$, $b < 0$ and $c > 0$. The graph of the equation in the xy-plane does NOT pass through which quadrant?

A) Quadrant II

B) Quadrant III

C) Quadrant I

D) Quadrant IV

Answer: (A) Quadrant II

Solve for y: $y = -\frac{a}{b}x + \frac{c}{b}$. Since $b < 0$: slope $-\frac{a}{b} > 0$ and y-intercept $\frac{c}{b} < 0$. A line with positive slope and negative y-intercept passes through Quadrants I, III and IV and misses Quadrant II. Test: $a = 1, b = -1, c = 1$ gives $y = x - 1$.

Trap: Students forget that dividing by the negative b flips both signs, and read the slope as $-\frac{a}{b} < 0$.

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3. A-322 · Hard · Multiple choice · **No calculator needed**

For real numbers x and y, $x + y < 12$ and $x - y > 4$. Which of the following must be true?

A) $y > 4$

B) $x > 8$

C) $y < 4$

D) $x < 8$

Answer: (C) $y < 4$

Rewrite the second inequality so both point the same way: $y - x < -4$. Adding $x + y < 12$ and $y - x < -4$ gives $2y < 8$, so $y < 4$. The value of x is not bounded: $(100, -90)$ satisfies both.

Trap: Students subtract the inequalities directly ($x + y - (x - y) < 12 - 4$) or add them as written to get a false bound on x.

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4. A-326 · Hard · SPR · **No calculator needed**

In the xy-plane, the region with $x \geq 0$, $y \geq 0$ and $2x + y \leq 12$ is a triangle. The line $y = mx$ divides this triangle into two regions of equal area. What is the value of m?

Answer: 2

The vertices are $(0,0)$, $(6,0)$ and $(0,12)$. A line through the vertex $(0,0)$ halves the area exactly when it passes through the midpoint of the opposite side, $(3,6)$. So $m = \frac{6}{3} = 2$. Check: the piece below the line, with vertices $(0,0)$, $(6,0)$ and $(3,6)$, has area $\frac{1}{2} \cdot 6 \cdot 6 = 18$, half of 36 ✓.

Trap: Students halve a leg (for example the line through $(3,0)$ or $(0,6)$) instead of using the midpoint of the side opposite the origin.

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5. A-331 · Hard · Multiple choice · **No calculator needed**

If $7x - 2y = 3x + 6y$ and x and y are nonzero, what is the value of $\frac{x+y}{x-y}$?

A) The value cannot be determined from the information given.

B) $\frac{1}{3}$

C) 2

D) 3

Answer: (D) 3

$4x = 8y$, so $x = 2y$. Then $\frac{x+y}{x-y} = \frac{3y}{y} = 3$.

Trap: Students believe one equation in two unknowns cannot fix a value, and choose “cannot be determined”.

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6. A-333 · Hard · Multiple choice · **No calculator needed**

A wall is $9x$ meters long, where x is a positive integer. Lamps are placed at both ends of the wall and every $\frac{9}{7}$ meters along it. Which expression gives the number of lamps?

A) $\frac{81x}{7} + 1$

B) $7x$

C) $7x + 1$

D) $9x + 1$

Answer: (C) $7x + 1$

Number of gaps $= \frac{9x}{9/7} = 7x$. With a lamp at both ends, lamps $=$ gaps $+1 = 7x + 1$. Check $x = 7$: 63 m, 49 gaps, 50 lamps.

Trap: Students count gaps instead of lamps ($7x$), or divide the wrong way ($\frac{81x}{7} + 1$).

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7. A-337 · Hard · Multiple choice · **No calculator needed**

The solution to the system $ax + by = 31$ and $bx + ay = 19$, where a and b are constants, is $(4,1)$. What is the value of $a^2 - b^2$?

A) 16

B) 40

C) 10

D) 58

Answer: (B) 40

$4a + b = 31$ and $a + 4b = 19$. Adding: $5a + 5b = 50$, so $a + b = 10$. Subtracting: $3a - 3b = 12$, so $a - b = 4$. $a^2 - b^2 = (a+b)(a-b) = 40$ ($a = 7$, $b = 3$).

Trap: Students report $a + b = 10$, $(a-b)^2 = 16$ or $a^2 + b^2 = 58$.

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8. A-339 · Hard · Multiple choice · **No calculator needed**

For every real number m, the graph of $y = mx + 3m - 2$ in the xy-plane passes through the point P. What are the coordinates of P?

A) $(-3, -2)$

B) $(0, -2)$

C) $(3, -2)$

D) $(-2, 3)$

Answer: (A) $(-3, -2)$

Rewrite: $y + 2 = m(x + 3)$. This holds for every m only when $x + 3 = 0$ and $y + 2 = 0$, so $P = (-3, -2)$.

Trap: Students test only $m = 0$ and pick $(0, -2)$, or make a sign error.

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9. A-342 · Hard · Multiple choice · **No calculator needed**

For how many integer values of k does the point (k, k) satisfy both $y \geq 2x - 5$ and $y > -x + 1$?

A) 5

B) 4

C) Infinitely many

D) 6

Answer: (A) 5

$k \geq 2k - 5 \Rightarrow k \leq 5$. $k > -k + 1 \Rightarrow k > \frac{1}{2}$. Integers $1, 2, 3, 4, 5$: 5 values.

Trap: Students treat $\geq$ as $>$ (getting 4) or use only one inequality.

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10. A-324 · Medium · Multiple choice · **No calculator needed**

A farm has 40 hectares of date palms and 25 hectares of olive trees. The equation $40x + 25y = 3{,}600$ represents the farm's total harvest, in tonnes, last season. What does x represent?

A) The average date harvest per hectare, in tonnes

B) The number of tonnes harvested per date palm tree

C) The number of hectares of date palms

D) The total date harvest, in tonnes

Answer: (A) The average date harvest per hectare, in tonnes

40 is the number of date-palm hectares, so $40x$ is the date harvest in tonnes; x is the date harvest per hectare, in tonnes.

Trap: Because x is usually a count, students choose "the number of hectares", missing that the coefficients are the counts here.

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11. A-327 · Medium · Multiple choice · **No calculator needed**

The function $V(t) = 1{,}200 - 0.75t$ gives the volume, in liters, of water in a tank t minutes after it starts draining. Which function gives the volume, in liters, h hours after the tank starts draining?

A) $1{,}200 - 0.75(h + 60)$

B) $1{,}200 - 45h$

C) $20 - 0.75h$

D) $1{,}200 - 0.0125h$

Answer: (B) $1{,}200 - 45h$

$t = 60h$, so $V = 1{,}200 - 0.75(60h) = 1{,}200 - 45h$.

Trap: Students divide the rate by 60 instead of multiplying, or shift the input.

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12. **A-330** · Medium · SPR · **No calculator needed**

A shopper spends exactly 600 dollars on x kilograms of oranges at 4 dollars per kilogram and y kilograms of apples at 5 dollars per kilogram, so $4x + 5y = 600$. If the shopper buys 10 more kilograms of oranges and still spends 600 dollars, by how many kilograms does y decrease?

Answer: 8

$4\Delta x + 5\Delta y = 0 \Rightarrow 4(10) + 5\Delta y = 0 \Rightarrow \Delta y = -8$: y decreases by 8 kg.

Trap: Students answer -10 (one-for-one trade) or -12.5 (dividing by 4 and multiplying by 5).

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13. **A-341** · Medium · Multiple choice · **No calculator needed**

Which inequality has the solution set $x \leq -3$ or $x \geq 11$?

A) $|x + 4| \geq 7$

B) $|x - 4| \leq 7$

C) $|x - 4| \geq 7$

D) $|x - 7| \geq 4$

Answer: (C) $|x - 4| \geq 7$

The centre of -3 and 11 is 4 and the distance to each is 7; the solutions lie outside: $|x - 4| \geq 7$.

Trap: Students reverse the inequality, flip the sign of the centre, or swap centre and distance.

Advanced Math (14 items)

14. **AM-310** · Hard · Multiple choice · **Desmos**

How many real solutions does the equation $x^2 = 2^x$ have?

A) 1

B) 2

C) 3

D) 4

Answer: (C) 3

$x = 2$ and $x = 4$ work by inspection. For $x < 0$, x^2 grows to ∞ as $x \to -\infty$ while $2^x \to 0$, and at $x = 0$, $x^2 = 0 < 1 = 2^0$, so the graphs also cross once for $x < 0$ (near $x \approx -0.77$). Total 3. Desmos: graph $y = x^2$ and $y = 2^x$ and zoom out.

Trap: Students find 2 and 4 by testing integers and miss the negative root.

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15. **AM-326** · Hard · Multiple choice · **Desmos**

What is the sum of all distinct real solutions of $x^3 - 2x^2 - 15x + 36 = 0$?

A) 2

B) 7

C) -2

D) -1

Answer: (D) -1

$x^3 - 2x^2 - 15x + 36 = (x - 3)^2(x + 4)$. The distinct solutions are 3 and -4; sum $= -1$. (Vieta's sum, 2, counts the double root twice.) Desmos: the graph touches the x-axis at 3 and crosses at -4.

Trap: Students use the sum of roots from the coefficients (2) without checking for a repeated root.

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16. AM-304 · Hard · SPR · **No calculator needed**

The function g is defined by $g(x) = \dfrac{x^2 - kx - 18}{x + 3}$, where k is a constant. The graph of $y = g(x)$ in the xy-plane is a line with exactly one point removed. What is the y-coordinate of the removed point?

Answer: -9

The graph is a line only if $x + 3$ cancels, so $x = -3$ is a zero of the numerator: $9 + 3k - 18 = 0$, $k = 3$. Then $x^2 - 3x - 18 = (x + 3)(x - 6)$ and $g(x) = x - 6$ for $x \neq -3$. The removed point is $(-3, -9)$.

Trap: Students give the x-coordinate -3, or answer 0 because the function is undefined there.

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17. AM-306 · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x) = x^2 - 9$ for $x < 2$ and $h(x) = -2x + 10$ for $x \geq 2$. What is the sum of all values of m for which $h(m) = 0$?

A) 0

B) 2

C) 5

D) 8

Answer: (B) 2

For $x < 2$: $x^2 = 9$ gives $x = \pm 3$, but 3 is not less than 2, so only -3. For $x \geq 2$: $x = 5$ ✓. Sum $= -3 + 5 = 2$.

Trap: Students keep $x = 3$ although it is outside its piece, giving 5 or 8.

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18. AM-308 · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = x^2 - 1$. How many distinct real solutions does the equation $f(f(x)) = 0$ have?

A) 1

B) 4

C) 2

D) 3

Answer: (D) 3

$f(u) = 0$ when $u = \pm 1$. $f(x) = 1 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$. $f(x) = -1 \Rightarrow x^2 = 0 \Rightarrow x = 0$ (one solution only). Total: 3.

Trap: Students expect $2 \times 2 = 4$ solutions, or stop at the zeros of f and answer 2.

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19. AM-313 · Hard · Multiple choice · **No calculator needed**

A kiosk's daily profit, in hundreds of dollars, is modeled by $P(x) = -2x^2 + 12x + 5$, where x is the price of an item in dollars. The model is used only for $4 \leq x \leq 7$. What is the greatest daily profit, in hundreds of dollars, predicted by the model on this interval?

A) 21

B) 23

C) -9

D) 5

Answer: (A) 21

The vertex is at $x = -\dfrac{12}{2(-2)} = 3$, outside the interval. On $4 \leq x \leq 7$ the parabola is decreasing, so the maximum is $P(4) = -32 + 48 + 5 = 21$.

Trap: Students use the vertex value $P(3) = 23$ without checking the interval.

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20. AM-315 · Hard · Multiple choice · **No calculator needed**

When the polynomial $p(x)$ is divided by $x-5$, the remainder is 8. What is the remainder when $p(2x-1)$ is divided by $x-3$?

A) 15

B) 8

C) 5

D) 3

Answer: (B) 8

By the remainder theorem, the remainder is the value of $p(2x-1)$ at $x=3$, which is $p(5)=8$.

Trap: Students transform the remainder ($2\cdot 8-1=15$) or answer the input 5.

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21. AM-319 · Hard · SPR · **No calculator needed**

The function f is defined by $f(x)=a\big((x-3)^2-b\big)\big((x-3)^2-c\big)$, where a, b and c are constants. If $f(1)=40$ and $f(6)=-14$, what is the value of $f(5)+f(0)$?

Answer: 26

x appears only in $(x-3)^2$, so the graph is symmetric about $x=3$. $f(5)$ uses $(5-3)^2=4$, the same as $f(1)$, so $f(5)=40$. $f(0)$ uses 9, the same as $f(6)$, so $f(0)=-14$. Sum $=26$.

Trap: Students try to solve for a, b and c (three unknowns, two equations) and decide it cannot be done.

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22. AM-321 · Hard · Multiple choice · **No calculator needed**

In the equation $\dfrac{x^2}{\sqrt{x^2-k^2}}=\dfrac{k^2}{\sqrt{x^2-k^2}}+24$, k is a positive constant. Which of the following is a solution?

A) $-\sqrt{k^2+576}$

B) $\sqrt{576-k^2}$

C) $\sqrt{k^2+24}$

D) $k+24$

Answer: (A) $-\sqrt{k^2+576}$

$\dfrac{x^2-k^2}{\sqrt{x^2-k^2}}=24\Rightarrow\sqrt{x^2-k^2}=24\Rightarrow x^2=k^2+576\Rightarrow x=\pm\sqrt{k^2+576}$. Only the negative root is offered. Check $k=7$: $x=-25$.

Trap: Students forget to square 24 (choosing $\sqrt{k^2+24}$) or reject a negative answer.

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23. AM-323 · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x)=3x^2-5x+2$. Which expression is equivalent to $f(a+1)-f(a)$?

A) 0

B) $6a-2$

C) $6a+3$

D) $6a+8$

Answer: (B) $6a-2$

$f(a+1)=3(a+1)^2-5(a+1)+2=3a^2+a$. $f(a)=3a^2-5a+2$. Difference: $6a-2$.

Trap: Students assume $f(a+1)=f(a)+f(1)$, giving 0 because $f(1)=0$, or drop the linear term.

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24. **AM-325** · Hard · Multiple choice · **No calculator needed**

The equations $x^2 - x - 12 = 0$ and $x^2 + kx + 8 = 0$ have a common solution, where k is a positive constant. What is the value of k?

A) -6

B) $\frac{17}{3}$

C) $-\frac{17}{3}$

D) 6

Answer: (B) $\frac{17}{3}$

$x^2-x-12=(x-4)(x+3)$. Common root 4: $16+4k+8=0$, $k=-6$ (not positive). Common root -3: $9-3k+8=0$, $k=\frac{17}{3}$.

Trap: Students test only the first root and answer -6, or ignore the condition $k>0$.

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25. **AM-328** · Hard · Multiple choice · **No calculator needed**

The functions f and g are defined by $f(x) = 7 \cdot 4^x$ and $g(x) = f(x - 1.5)$. For all values of x, $g(x) = k \cdot f(x)$, where k is a constant. What is the value of k?

A) -6

B) $\frac{7}{8}$

C) $\frac{1}{8}$

D) 8

Answer: (C) $\frac{1}{8}$

$g(x) = 7 \cdot 4^{x-1.5} = 7 \cdot 4^x \cdot 4^{-1.5} = f(x) \cdot \frac{1}{8}$. So $k = \frac{1}{8}$.

Trap: Students shift in the wrong direction ($k = 8$) or subtract 1.5 from the base.

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26. **AM-330** · Hard · Multiple choice · **No calculator needed**

The function g is defined by $g(x) = 3^x$. If $g(a) = 4$, what is the value of $g(2a - 1)$?

A) $\frac{16}{3}$

B) 15

C) $\frac{8}{3}$

D) 7

Answer: (A) $\frac{16}{3}$

$g(2a-1) = 3^{2a-1} = \frac{(3^a)^2}{3} = \frac{16}{3}$.

Trap: Students apply the input operations to the output ($2 \cdot 4 - 1 = 7$) or compute $4^2 - 1$.

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27. **AM-318** · Medium · SPR · **No calculator needed**

How many integers x satisfy the inequality $x(x - 4) < 2x$?

Answer: 5

$x^2 - 6x < 0 \Rightarrow x(x-6) < 0 \Rightarrow 0 < x < 6$. Integers: $1, 2, 3, 4, 5$, so 5.

Trap: Students divide both sides by x (not allowed when x may be negative or zero) and get $x < 6$, which has infinitely many integers.

Geometry and Trigonometry (22 items)

28. GT-253 · Hard · Multiple choice · **Calculator**

Square pyramids X and Y have the same base, a square with side length 10. Pyramid X has height 12 and pyramid Y has height 24. The total surface area of Y is approximately what percent greater than the total surface area of X?

A) 89%

B) 100%

C) 64%

D) 50%

Answer: (C) 64%

Slant heights: X: $\sqrt{12^2+5^2}=13$; Y: $\sqrt{24^2+5^2}=\sqrt{601}\approx 24.52$. Areas: $X=100+20(13)=360$; $Y=100+20(24.52)\approx 590.3$. $\dfrac{590.3}{360}\approx 1.64$: about 64% greater.

Trap: Students expect doubling (100%), use the height as the slant height, or compare lateral areas only (89%).

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29. GT-250 · Hard · SPR · **Desmos**

In the equation $\sin\left(\dfrac{39\pi}{p}\right)=\dfrac{\sqrt{3}}{2}$, p is a constant and $p>60$. What is the value of p?

Answer: 117

$p>60$ means $0<\dfrac{39\pi}{p}<\dfrac{39\pi}{60}=0.65\pi$. The only angle in this interval with sine $\dfrac{\sqrt{3}}{2}$ is $\dfrac{\pi}{3}$ (the next, $\dfrac{2\pi}{3}$, gives $p=58.5$). So $\dfrac{39}{p}=\dfrac{1}{3}$, $p=117$.

Trap: Students take $\dfrac{2\pi}{3}$ and get 58.5, ignoring the condition, or work in degrees.

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30. GT-233 · Hard · SPR · **No calculator needed**

The graph of $x^2+y^2+10x-8y+k=0$ in the xy-plane is a circle, where k is a constant. What is the greatest possible integer value of k?

Answer: 40

Complete the squares: $(x+5)^2+(y-4)^2=41-k$. A circle needs $r^2>0$, so $k<41$. Greatest integer: 40. ($k=41$ gives a single point.)

Trap: Students allow $r^2=0$ and answer 41, or reverse the inequality.

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31. GT-235 · Hard · Multiple choice · **No calculator needed**

Points A, B and C lie on a circle with radius 12, and C lies on the major arc AB. The measure of angle ACB is $\dfrac{2\pi}{9}$ radians. What is the length of minor arc AB?

A) $\dfrac{16\pi}{3}$

B) $\dfrac{8\pi}{3}$

C) $\dfrac{64\pi}{3}$

D) $\dfrac{56\pi}{3}$

Answer: (A) $\dfrac{16\pi}{3}$

The central angle is twice the inscribed angle: $\dfrac{4\pi}{9}$. Arc length $=12\cdot\dfrac{4\pi}{9}=\dfrac{16\pi}{3}$.

Trap: Students use the inscribed angle as the central angle and get $\dfrac{8\pi}{3}$.

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32. GT-237 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle, $AC = 16$ and $BC = 12$. Point E lies on $\overline{AC}$ with $AE = 10$, and point F lies on $\overline{AB}$ so that $\overline{EF} \perp \overline{AB}$. What is the length of $\overline{EF}$?

A) 8

B) 6

C) 7.5

D) 12.5

Answer: (B) 6

$AB = 20$. Triangles AFE and ACB are similar (shared angle A, right angles at F and C), with hypotenuses AE and AB. So $\dfrac{EF}{BC} = \dfrac{AE}{AB}$: $EF = 12 \cdot \dfrac{10}{20} = 6$.

Trap: Students match AE with the leg AC ($EF = 12 \cdot \frac{10}{16} = 7.5$) or give $AF = 8$.

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33. GT-240 · Hard · Multiple choice · **No calculator needed**

A right circular cone stands with its vertex pointing up. Its base radius is 6 and its height is 12. Water fills the cone to a depth of 6, measured from the base. What is the volume of the water?

A) 18π

B) 72π

C) 108π

D) 126π

Answer: (D) 126π

The empty top part is a cone similar to the whole with scale $\frac{1}{2}$, so its volume is $\frac{1}{8}$ of the cone. Cone volume $= \frac{1}{3}\pi(36)(12) = 144\pi$. Water $= \frac{7}{8} \cdot 144\pi = 126\pi$.

Trap: Students take the water as the small cone (18π) or halve the volume (72π).

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34. GT-242 · Hard · Multiple choice · **No calculator needed**

Line ℓ passes through $(0, 0)$ and $(a, 12)$, where $a > 0$. The acute angle between ℓ and the positive y-axis measures $60°$. What is the value of a?

A) $12\sqrt{3}$

B) $4\sqrt{3}$

C) 24

D) 6

Answer: (A) $12\sqrt{3}$

The angle with the positive x-axis is $90° - 60° = 30°$, so $\tan 30° = \dfrac{12}{a}$ and $a = \dfrac{12}{1/\sqrt{3}} = 12\sqrt{3}$.

Trap: Students use $60°$ as the angle with the x-axis and get $a = 4\sqrt{3}$.

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35. GT-244 · Hard · Multiple choice · **No calculator needed**

A square has one side on the diameter of a semicircle of radius 10, and its other two vertices lie on the arc. What is the area of the square?

A) 100

B) 50

C) 80

D) 200

Answer: (C) 80

Let the side be s. A top vertex is $\frac{s}{2}$ from the centre horizontally and s up, so $s^2 + \left(\frac{s}{2}\right)^2 = 100$, $\frac{5}{4}s^2 = 100$, $s^2 = 80$.

Trap: Students use the full-circle result (area $2r^2 = 200$) or half of it.

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36. GT-247 · Hard · Multiple choice · **No calculator needed**

In a circle of radius 13, two parallel chords have lengths 24 and 10. Which of the following could be the distance between the two chords?

A) 12

B) 17

C) 26

D) 5

Answer: (B) 17

Distances from the centre: $\sqrt{169-144}=5$ and $\sqrt{169-25}=12$. Same side of the centre: $12-5=7$; opposite sides: $12+5=17$. Only 17 is offered.

Trap: Students picture only one case, or add the half-chord lengths.

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37. GT-251 · Hard · Multiple choice · **No calculator needed**

A rectangle is inscribed in a circle with area 676π. The length of the rectangle's diagonal is twice the length of its shorter side. What is the perimeter of the rectangle?

A) $52+52\sqrt{3}$

B) 104

C) $26+26\sqrt{3}$

D) $104+52\sqrt{3}$

Answer: (A) $52+52\sqrt{3}$

$r=26$, so the diagonal (a diameter) is 52 and the short side is 26. The diagonal is twice the short side, so the triangle is $30°$–$60°$–$90°$ and the long side is $26\sqrt{3}$. Perimeter $=52+52\sqrt{3}$.

Trap: Students assume a square, or use the radius as the diagonal.

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38. GT-254 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $18°$. Which additional fact is NOT sufficient to determine the measure of angle B?

A) $m\angle B = m\angle C$

B) $m\angle C = 3\,m\angle B$

C) Triangle ABC is isosceles.

D) $m\angle C = 90°$

Answer: (C) Triangle ABC is isosceles.

If the triangle is isosceles, angle A could be a base angle or the vertex angle, so B could be $18°$, $81°$ or $144°$. Each other fact fixes B: $B+3B=162 \Rightarrow B=40.5°$; $B=C \Rightarrow B=81°$; $C=90° \Rightarrow B=72°$.

Trap: Students think "isosceles" fixes the angles and stop at the first case.

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39. GT-257 · Hard · Multiple choice · **No calculator needed**

In triangle JKL, the measure of angle J is $(ab)°$ and the measure of angle K is $(cd)°$, where $ab+cd=90$. If $\sin J = 0.6$, what is the value of $\tan K$?

A) $\dfrac{3}{4}$

B) $\dfrac{4}{3}$

C) $\dfrac{4}{5}$

D) The value cannot be determined from the information given.

Answer: (B) $\dfrac{4}{3}$

$J + K = 90°$, so angle $L = 90°$ and J, K are complementary: $\cos K = \sin J = 0.6$, $\sin K = 0.8$, $\tan K = \dfrac{0.8}{0.6} = \dfrac{4}{3}$.

Trap: Students decide there is not enough information because a, b, c, d are unknown.

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40. GT-259 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, $AB = AC = 17$ and $BC = 16$. What is the value of $\tan B$?

A) $\dfrac{8}{15}$

B) $\dfrac{15}{8}$

C) $\dfrac{15}{17}$

D) $\dfrac{16}{17}$

Answer: (B) $\dfrac{15}{8}$

The altitude from A bisects $\overline{BC}$: half-base 8, altitude $\sqrt{17^2 - 8^2} = 15$. $\tan B = \dfrac{15}{8}$.

Trap: Students use the whole base, flip the ratio, or give $\sin B$.

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41. GT-261 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle and $\cos A = k$. Which expression is equal to $\sin^2 A + \sin^2 B$?

A) 1

B) The value cannot be determined from the information given.

C) $2 - 2k^2$

D) $2k^2$

Answer: (A) 1

A and B are complementary, so $\sin B = \cos A$. $\sin^2 A + \sin^2 B = \sin^2 A + \cos^2 A = 1$.

Trap: Students think the value depends on k or cannot be found.

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42. GT-263 · Hard · Multiple choice · **No calculator needed**

A straight railway track up a hill has a constant grade of 75%, meaning it rises 75 meters for every 100 meters of horizontal distance. The track is 300 meters long, measured along the slope. What is the vertical rise of the track, in meters?

A) 400

B) 240

C) 180

D) 225

Answer: (C) 180

Grade is $\tan\theta = \dfrac{3}{4}$, so the sides are in ratio $3 : 4 : 5$ and $\sin\theta = \dfrac{3}{5}$. Rise $= 300 \cdot \dfrac{3}{5} = 180$.

Trap: Students treat the grade as the sine ($300 \cdot 0.75 = 225$) or give the horizontal run (240).

---

43. GT-265 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle B is $45°$ and the measure of angle C is $30°$. The altitude from A to $\overline{BC}$ has length 6. What is the length of $\overline{BC}$?

A) 12

B) $6 + 2\sqrt{3}$

C) $12 + 6\sqrt{2}$

D) $6 + 6\sqrt{3}$

Answer: (D) $6+6\sqrt{3}$

The altitude meets $\overline{BC}$ at D. In the $45°$ triangle, $BD = 6$. In the $30°$ triangle, $DC = \dfrac{6}{\tan 30°} = 6\sqrt{3}$. $BC = 6+6\sqrt{3}$.

Trap: Students multiply by $\tan 30°$ instead of dividing $(6+2\sqrt{3})$.

---

44. GT-268 · Hard · Multiple choice · **No calculator needed**

From the top of a cliff 45 meters high, the angles of depression to two boats on opposite sides of the cliff are $30°$ and $45°$. The boats and the base of the cliff are on a straight line. What is the distance, in meters, between the boats?

A) $45\sqrt{3}-45$

B) $45+15\sqrt{3}$

C) $45\sqrt{3}$

D) $45+45\sqrt{3}$

Answer: (D) $45+45\sqrt{3}$

Horizontal distances: $\dfrac{45}{\tan 30°} = 45\sqrt{3}$ and $\dfrac{45}{\tan 45°} = 45$. Opposite sides, so add: $45+45\sqrt{3}$.

Trap: Students subtract the distances (the same-side case) or multiply by $\tan 30°$.

---

45. GT-270 · Hard · SPR · **No calculator needed**

A rectangle has side lengths 6 and $6\sqrt{3}$. What is the measure, in degrees, of the acute angle formed by its diagonals?

Answer: 60

Each diagonal makes an angle with the longer side whose tangent is $\dfrac{6}{6\sqrt{3}} = \dfrac{1}{\sqrt{3}}$, so $30°$. The triangle formed by the longer side and the two half-diagonals has base angles $30°$ and apex angle $120°$. The adjacent angle at the centre, facing the shorter side, is $180° - 120° = 60°$.

Trap: Students give the diagonal-to-side angle $(30°)$ or the obtuse angle $(120°)$.

---

46. GT-272 · Hard · Multiple choice · **No calculator needed**

Parallel lines ℓ and m are horizontal, with ℓ above m. Point P lies between the lines. Point A on ℓ and point B on m are both to the right of P. Segment $\overline{PA}$ makes a $38°$ angle with line ℓ, and segment $\overline{PB}$ makes a $47°$ angle with line m, both angles opening toward P. What is the measure, in degrees, of angle APB?

A) 42.5

B) 95

C) 85

D) 9

Answer: (C) 85

Draw a line through P parallel to ℓ and m. Alternate interior angles give $38°$ above it and $47°$ below it, so $\angle APB = 38° + 47° = 85°$.

Trap: Students compute $180 - (38+47) = 95$ or the difference 9.

---

47. GT-248 · Medium · Multiple choice · **Calculator**

The radius of a right circular cylinder is increased by 20% and its height is decreased by 25%. What is the effect on the volume of the cylinder?

A) It does not change.

B) It increases by 8%.

C) It decreases by 5%.

D) It increases by 19%.

Answer: (B) It increases by 8%.

V scales by $(1.2)^2(0.75) = 1.08$: an increase of 8%.

Trap: Students add percents ($20 - 25 = -5$) or forget to square the radius factor.

---

48. GT-245 · Medium · Multiple choice · **No calculator needed**

In the xy-plane, the endpoints of a diameter of a circle are $(-2, 3)$ and $(6, -1)$. Which equation represents the circle?

A) $x^2 + y^2 - 4x - 2y - 15 = 0$

B) $x^2 + y^2 - 4x - 2y - 20 = 0$

C) $x^2 + y^2 - 4x - 2y - 75 = 0$

D) $x^2 + y^2 + 4x + 2y - 15 = 0$

Answer: (A) $x^2 + y^2 - 4x - 2y - 15 = 0$

Centre $(2, 1)$, $r^2 = (6-2)^2 + (-1-1)^2 = 20$. $(x-2)^2 + (y-1)^2 = 20 \Rightarrow x^2 + y^2 - 4x - 2y + 5 = 20 \Rightarrow x^2 + y^2 - 4x - 2y - 15 = 0$.

Trap: Students use the diameter squared (80) as r^2 or drop the constants $4 + 1$ when expanding.

---

49. GT-274 · Medium · Multiple choice · **No calculator needed**

In triangle ABC, $AB = 2x + 1$, $BC = x + 6$ and $CA = 3x - 4$. The perimeter of the triangle is 45. Which angle of the triangle has the greatest measure?

A) Angle A

B) It cannot be determined from the information given.

C) Angle C

D) Angle B

Answer: (D) Angle B

$6x + 3 = 45 \Rightarrow x = 7$: $AB = 15$, $BC = 13$, $CA = 17$. The greatest angle is opposite the longest side $\overline{CA}$, which is angle B.

Trap: Students name a vertex of the longest side (A or C) instead of the opposite angle.

Problem-Solving and Data Analysis (27 items)

50. PS-166 · Hard · Multiple choice · **Calculator**

District A has $120{,}000$ residents and a population density of $4{,}000$ residents per square kilometer. District B has $60{,}000$ residents and a population density of $1{,}500$ residents per square kilometer. Which is closest to the population density, in residents per square kilometer, of the two districts combined?

A) $5{,}500$

B) $3{,}167$

C) $2{,}571$

D) $2{,}750$

Answer: (C) $2{,}571$

Areas: $\frac{120{,}000}{4{,}000} = 30$ and $\frac{60{,}000}{1{,}500} = 40$ km^2. Combined density $= \frac{180{,}000}{70} \approx 2{,}571$.

Trap: Students average the densities ($2{,}750$) or weight them by population ($\approx 3{,}167$).

---

51. PS-178 · Hard · Multiple choice · **Calculator**

A store's profit on each jacket is 37.5% of the selling price. The profit is what percent of the cost of the jacket to the store?

A) 37.5%

B) 27.3%

C) 62.5%

D) 60%

Answer: (D) 60%

Let the price be 100: profit 37.5, cost 62.5. $\dfrac{37.5}{62.5} = 0.6 = 60\%$.

Trap: Students keep the selling price as the base (37.5%) or compute cost as a percent of price (62.5%).

---

52. PS-181 · Hard · Multiple choice · **Calculator**

A worker spends 30% of her monthly salary on rent, then 25% of the remaining amount on food, then 40% of what is left after that on savings. She has $3{,}780$ dollars left. What is her monthly salary, in dollars?

A) $75{,}600$

B) $7{,}200$

C) $12{,}000$

D) $6{,}300$

Answer: (C) $12{,}000$

Remaining fraction: $0.7 \cdot 0.75 \cdot 0.6 = 0.315$. Salary $= \dfrac{3{,}780}{0.315} = 12{,}000$.

Trap: Students add the percents (95% spent) or skip a step.

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53. PS-192 · Hard · Multiple choice · **Calculator**

A bank sells US dollars at 3.77 riyals per dollar. It first deducts a 1% commission from the riyals a customer pays. How many US dollars does a customer receive for $5{,}655$ riyals?

A) $1{,}515$

B) $1{,}485$

C) $21{,}319$

D) $1{,}500$

Answer: (B) $1{,}485$

Riyals exchanged: $0.99 \cdot 5{,}655 = 5{,}598.45$. Dollars $= \dfrac{5{,}598.45}{3.77} = 1{,}485$.

Trap: Students ignore the commission ($1{,}500$), add it, or multiply by the rate.

---

54. PS-159 · Hard · SPR · **No calculator needed**

The data set $12, 15, 19, 24, 30$ has a range of 18. A sixth value, x, is added, and the range of the new data set is 1.5 times the original range. What is the sum of all possible values of x?

Answer: 42

New range $= 27$. Either x is the new maximum: $x - 12 = 27$, $x = 39$; or the new minimum: $30 - x = 27$, $x = 3$. Sum $= 42$.

Trap: Students assume the new value is the maximum and answer 39.

---

55. PS-161 · Hard · SPR · **No calculator needed**

A data set contains the values $1, 2, 3, 4, 5$ with frequencies $4, k, 6, 3, 5$, respectively, where k is a positive integer. The median of the data set is 3. What is the greatest possible value of k?

Answer: 9

There are $18 + k$ values, and $4 + k$ of them are at most 2. For $k = 9$: 27 values, the 14th is 3 ✓. For $k = 10$: 28 values, the 14th is 2 and the 15th is 3, so the median is 2.5. Greatest $k = 9$.

Trap: Students ignore that the total changes with k, or forget that an even count averages the two middle values.

---

56. PS-163 · Hard · Multiple choice · **No calculator needed**

Data set A has 11 values with median 40. Data set B has 11 values with median 52. The two data sets are combined. Which of the following could be the median of the combined 22 values?

A) 46

B) 38

C) 60

D) 54

Answer: (A) 46

At most 5 values of A and at most 5 values of B are less than 40, so at most 10 combined values are below 40 and the 11th value is at least 40. Likewise at most 10 combined values are greater than 52, so the 12th value is at most 52. The median lies between 40 and 52; only 46 is in this range.

Trap: Students think nothing can be said, or believe the median must equal the average of the two medians.

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57. PS-168 · Hard · SPR · **No calculator needed**

In a school, a student is chosen at random. The probability that the student is in grade 10, given that the student is in the robotics club, is $\frac{3}{8}$. The probability that the student is in the robotics club, given that the student is in grade 10, is $\frac{1}{4}$. There are 96 grade 10 students. How many robotics club members are NOT in grade 10?

Answer: 40

Grade 10 robotics members $= \frac{1}{4} \cdot 96 = 24$. These are $\frac{3}{8}$ of the club, so the club has $\frac{24}{3/8} = 64$ members. Not in grade 10: $64 - 24 = 40$.

Trap: Students use the same denominator for both conditional probabilities.

---

58. PS-170 · Hard · Multiple choice · **No calculator needed**

In grade 10, 18 students are in the robotics club and 42 are not. In grade 11, 30 students are in the robotics club and 60 are not. A student is chosen at random from the students who are in grade 10 OR in the robotics club. What is the probability that the student is in grade 11?

A) $\frac{1}{3}$

B) $\frac{1}{5}$

C) $\frac{1}{2}$

D) $\frac{5}{8}$

Answer: (A) $\frac{1}{3}$

The group is all 60 grade 10 students plus the 30 grade 11 robotics members: 90 students. Grade 11 students in it: 30. Probability $= \frac{30}{90} = \frac{1}{3}$.

Trap: Students use only the robotics club ($\frac{30}{48}$) or all students ($\frac{30}{150}$) as the denominator.

---

59. PS-172 · Hard · SPR · **No calculator needed**

The model $\hat{y} = 4.8x + 62$ estimates a town's population y, in thousands, where x is the number of decades since 1980. According to the model, what is the estimated increase in the population, in people, per year?

Answer: 480

4.8 thousand per decade $= 4{,}800$ people per 10 years $= 480$ people per year.

Trap: Students report 4.8, 48 or $4{,}800$ by converting only one of the units.

---

60. PS-175 · Hard · SPR · **No calculator needed**

Data set A has 10 integers: 4 are from 10 to 19 and 6 are from 20 to 29. Data set B has 10 integers: 3 are from 30 to 39 and 7 are from 40 to 49. What is the smallest possible value of (mean of B) $-$ (mean of A)?

Answer: 12

Make A as large as possible and B as small as possible. Greatest mean of $A = \dfrac{4(19)+6(29)}{10} = 25$. Least mean of $B = \dfrac{3(30)+7(40)}{10} = 37$. Smallest difference $= 12$.

Trap: Students use bin midpoints (difference 21) or think the means cannot be compared.

---

61. PS-180 · Hard · Multiple choice · **No calculator needed**

A club has 40 members, and 30% of the members are girls. How many girls must join the club, with no one leaving, so that girls make up 60% of the members?

A) 20

B) 12

C) 30

D) 28

Answer: (C) 30

Girls now: 12. $12 + g = 0.6(40 + g) \Rightarrow 0.4g = 12 \Rightarrow g = 30$.

Trap: Students take 60% of the old total ($24 - 12 = 12$) and forget the total grows.

---

62. PS-184 · Hard · Multiple choice · **No calculator needed**

Fresh dates are 80% water by mass. After drying, the dates are 50% water by mass. How many kilograms of dried dates are produced from 60 kilograms of fresh dates?

A) 30

B) 24

C) 42

D) 12

Answer: (B) 24

The non-water part does not change: $0.2 \cdot 60 = 12$ kg. After drying it is 50% of the mass: $\dfrac{12}{0.5} = 24$ kg.

Trap: Students subtract 30 percentage points of the mass ($60 \cdot 0.7 = 42$) or halve the mass.

---

63. PS-186 · Hard · Multiple choice · **No calculator needed**

At a school, 85% of the boys and 60% of the girls passed an exam, and 70% of all the students passed. What percent of the students at the school are boys?

A) 25%

B) 50%

C) 60%

D) 40%

Answer: (D) 40%

Let b be the fraction of boys: $0.85b + 0.6(1 - b) = 0.7 \Rightarrow 0.25b = 0.1 \Rightarrow b = 0.4 = 40\%$.

Trap: Students assume equal groups (50%) or swap the groups (60%).

---

64. PS-189 · Hard · Multiple choice · **No calculator needed**

A crew of 12 workers can finish a job in 20 days. After the crew has worked for 5 days, 3 more workers join, all working at the same rate. How many days in total does the job take?

A) 16

B) 25

C) 17

D) 12

Answer: (C) 17

Total work $= 12 \cdot 20 = 240$ worker-days. Done in 5 days: 60. Remaining 180 with 15 workers: 12 days. Total $5 + 12 = 17$.

Trap: Students give only the remaining days (12) or assume 15 workers from the start (16).

---

65. PS-191 · Hard · Multiple choice · **No calculator needed**

A printer prints p pages every m seconds. Which expression gives the number of minutes the printer takes to print n pages?

A) $\dfrac{60m}{pn}$

B) $\dfrac{60mn}{p}$

C) $\dfrac{pn}{60m}$

D) $\dfrac{mn}{60p}$

Answer: (D) $\dfrac{mn}{60p}$

Seconds per page $= \dfrac{m}{p}$; n pages take $\dfrac{mn}{p}$ seconds $= \dfrac{mn}{60p}$ minutes. Check $p = 10$, $m = 5$, $n = 120$: 60 s $= 1$ min.

Trap: Students multiply by 60 instead of dividing, or invert the rate.

---

66. PS-194 · Hard · Multiple choice · **No calculator needed**

A data set consists of $12, 15, 18, 21, 24, 27, 30, 33, 36, 40$. The value 40 is mistakenly recorded as 400. How do the median, the interquartile range (IQR) and the standard deviation of the recorded data compare with those of the correct data?

A) Only the median is unchanged.

B) The median, IQR and standard deviation are all unchanged.

C) The median is unchanged, but the IQR and standard deviation are greater.

D) The median and IQR are unchanged, but the standard deviation is greater.

Answer: (D) The median and IQR are unchanged, but the standard deviation is greater.

Only the maximum changed. The median (25.5) and the quartiles $(18$ and $33)$ do not involve the maximum, so the median and IQR are unchanged. The standard deviation measures distance from the mean and grows greatly (about 8.8 to about 113).

Trap: Students think every statistic changes with an outlier, or that the IQR is affected.

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67. PS-196 · Hard · Multiple choice · **No calculator needed**

At a company, the median monthly salary is $9{,}000$ dollars and the mean monthly salary is $14{,}500$ dollars. Which statement must be true?

A) The most common salary is $9{,}000$ dollars.

B) The standard deviation of the salaries is $5{,}500$ dollars.

C) No more than half of the employees earn more than $14{,}500$ dollars.

D) More than half of the employees earn more than $14{,}500$ dollars.

Answer: (C) No more than half of the employees earn more than $14{,}500$ dollars.

Any employee earning more than $14{,}500$ also earns more than the median $9{,}000$, and at most half of the employees earn more than the median. So no more than half earn more than $14{,}500$.

Trap: Students read the mean as the typical salary, or treat mean minus median as the standard deviation.

---

68. PS-198 · Hard · Multiple choice · **No calculator needed**

A data set has six values. Their deviations from 40 (value minus 40) are -7, 3, 12, -5, x and 2. The mean of the data set is 42. What is the value whose deviation from 40 is x?

A) 35

B) 7

C) 47

D) 52

Answer: (C) 47

The deviations from 40 sum to $6(42-40)=12$, not 0 (they sum to 0 only around the mean). $5+x=12$, $x=7$; the value is 47.

Trap: Students set the deviations' sum to 0 ($x=-5$, value 35) or stop at $x=7$.

---

69. PS-201 · Hard · Multiple choice · **No calculator needed**

The recorded values of a data set are $18, 22, 27, 30, 33, 36, 41, 47, 52$. The value 52 should have been 25. After the correction, how do the mean and the median change?

A) The mean decreases by 3 and the median is unchanged.

B) The median decreases and the mean is unchanged.

C) The mean decreases by 27 and the median decreases by 3.

D) The mean and the median both decrease by 3.

Answer: (D) The mean and the median both decrease by 3.

Mean: $34 \to 31$ (the sum drops by 27 over 9 values). Median: $33 \to 30$, because 25 moves below the middle and the middle value shifts. Both decrease by 3.

Trap: Students assume the median is unchanged when one value is corrected.

---

70. PS-203 · Hard · Multiple choice · **No calculator needed**

In an experiment, garden plots were randomly assigned to receive 0, 20 or 40 grams of a fertilizer. The plots given 40 grams had the highest mean yield. Which conclusion is best supported?

A) Doubling the fertilizer doubles the yield.

B) For plots like these, increasing the fertilizer from 0 to 40 grams caused higher yields.

C) No causal conclusion is possible.

D) Using 80 grams would give an even higher yield.

Answer: (B) For plots like these, increasing the fertilizer from 0 to 40 grams caused higher yields.

Random assignment supports a causal conclusion, but only within the tested doses and for plots like these.

Trap: Students extrapolate a valid causal result beyond the tested doses.

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71. PS-209 · Hard · Multiple choice · **No calculator needed**

For data set P, the line of best fit is $\hat{y}=4.2x+3$ and the correlation coefficient is 0.41. For data set Q, the line of best fit is $\hat{y}=0.6x+10$ and the correlation coefficient is 0.93. Which statement is true?

A) The associations cannot be compared because the slopes are different.

B) P has the stronger linear association because its slope is greater.

C) Q has a weaker association because its slope is less than 1.

D) Q has the stronger linear association.

Answer: (D) Q has the stronger linear association.

The strength of a linear association is measured by r, not by the slope. Q has r closer to 1, so its points fit a line more closely, relative to the spread of the data.

Trap: Students think a steeper slope means a stronger association.

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72. PS-177 · Medium · Multiple choice · **Calculator**

The monthly rent of an apartment increased by 60%. By what percent must the new rent decrease to return to the original rent?

A) 62.5%

B) 37.5%

C) 60%

D) 40%

Answer: (B) 37.5%

New rent $= 1.6R$. Decrease needed: $\dfrac{0.6R}{1.6R} = 0.375 = 37.5\%$.

Trap: Students reuse 60% or compute $100 - 60 = 40$.

---

73. PS-158 · Medium · SPR · **No calculator needed**

The daily high temperatures in a city for one week have a mean of $34°$C and a standard deviation of $4°$C. Each temperature is converted to degrees Fahrenheit using $F = 1.8C + 32$. What is the standard deviation of the converted temperatures, in degrees Fahrenheit?

Answer: 7.2

Multiplying every value by 1.8 multiplies the SD by 1.8; adding 32 shifts the values but does not change the spread. SD $= 1.8 \cdot 4 = 7.2$.

Trap: Students apply the full conversion ($1.8 \cdot 4 + 32 = 39.2$) or think the SD does not change.

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74. PS-200 · Medium · SPR · **No calculator needed**

In a data set, a value is called a low outlier if it is less than $Q_1 - 1.5 \cdot$ IQR, where Q_1 is the first quartile and IQR is the interquartile range. For a data set, $Q_1 = 28$ and the third quartile is 40. What is the greatest integer value that would be a low outlier?

Answer: 9

IQR $= 12$; fence $= 28 - 18 = 10$. A low outlier is strictly less than 10, so the greatest integer is 9.

Trap: Students include the boundary (10) or forget the factor 1.5 (16).

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75. PS-206 · Medium · Multiple choice · **No calculator needed**

To estimate the average weekly exercise time of adults in a city, a researcher randomly selected 400 people from the membership lists of the city's gyms. Why is the estimate likely to be biased?

A) The sample size is too small.

B) The selection was random, so the estimate is not biased.

C) The list leaves out adults who are not gym members.

D) There was no random assignment.

Answer: (C) The list leaves out adults who are not gym members.

The sampling frame (gym members) leaves out adults who are not gym members, who probably exercise less. Random selection from a biased list is still biased.

Trap: Students accept random selection without checking the list it came from.

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76. **PS-208** · Medium · Multiple choice · **No calculator needed**

For a data set, the correlation coefficient between height in centimeters and mass in kilograms is $r = 0.72$. The heights are converted to inches, the masses to pounds, and the two variables are swapped between the axes. What is the correlation coefficient for the converted data?

A) 0.72

B) -0.72

C) $0.72 \cdot \dfrac{2.2}{2.54}$

D) $\dfrac{1}{0.72}$

Answer: (A) 0.72

Correlation has no units and is symmetric: changing units (positive scale factors) and swapping the axes leave r unchanged, 0.72.

Trap: Students scale r by the conversion factors or take its reciprocal.
